

Supplementary Material

A. Calculation of the contribution of interstitials vs. vacancies

The following calculation demonstrated that the interstitial term on equation 16 of the main text can be considered of negligible influence respect the vacancy term, due to the difference of three orders of magnitude between them.

$$\frac{dV_{void}}{dt} = 4\pi R\Omega[D_v(C_v - C_v^V) - D_i C_i]$$

At $t=0.0001$, $C_v^V \approx 0$. For copper, we know that:

$$D_v = 3.9e^{-7} cm^2 s^{-1} [51]$$

$$C_v \approx 2e^{-4} cm^{-3} [36]$$

$$D_i = 3.19e^{-5} cm^2 s^{-1} [51]$$

$$C_i \approx 6e^{-10} cm^{-3} [36]$$

$$D_v C_v \approx 7.8e^{-11} \text{ and } D_i C_i \approx 1.9e^{-14}, \text{ therefore } D_v C_v \gg D_i C_i$$

B. Thermal diffusivity and stopping powers

First table shows the calculation of the stopping powers heavy ion/proton ratio for each of the elements that compose the GRCo-84 alloy. At the end of the table, a composition-averaged ratio is shown.

	Ei (eV)	7000000							
	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)	Ratio Stot	
Incoming	Cu	29 63.546	31201.4356	0.000167719	1.37543E-16	7.49208E-11	7.4921E-11		
Target	Cu	29 63.546							
	Ei (eV)	1630000							
	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Incoming	H	1 1.00794	2438733.63	3.04147E-06	3.67689E-21	3.83014E-12	3.8301E-12	19.5608631	
Target	Cu	29 63.546							
	Ei (eV)	7000000							
	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Incoming	Cu	29 63.546	34671.3284	0.000152461	1.16294E-16	7.49208E-11	7.4921E-11		
Target	Cr	24 51.9961							
	Ei (eV)	1630000							
	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Incoming	H	1 1.00794	3024802.35	2.4878E-06	3.12241E-21	3.83014E-12	3.8301E-12	19.5608575	
Target	Cr	24 51.9961							
	Ei (eV)	7000000							
	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Incoming	Cu	29 63.546	25167.6954	0.000203638	1.84162E-16	7.49208E-11	7.4921E-11		
Target	Nb	41 92.90638							
	Ei (eV)	1630000							
	Z	M	E	Sn(E)	Sn(Ei)	Se(Ei)	Stot(Ei)		
Incoming	H	1 1.00794	1640426.02	4.40062E-06	4.89236E-21	3.83014E-12	3.8301E-12	19.5608753	
Target	Nb	41 92.90638							
88/8/4 averaged composition								19.5608631	

S 1. Calculation of the stopping powers heavy ion/proton ratio for each of the elements that compose the GRCo-84 alloy. At the end of the table, a composition-averaged ratio is shown. Excel file is accessible at the repository of this paper [3].

This second table shows the calculated drop in thermal diffusivity for Cu-ion irradiation and proton irradiation, and the calculated ratio.

Cu ions 3+ Low rate DPA		
Beam	SAWf	Th.diff
before	5.57E+08	6.77E-06
after	5.55E+08	6.60E-06
off/on	-0.40	-2.46
Protons at Low rate DPA		
Beam	SAWf	Th.diff
before	5.54E+08	6.59E-06
after	5.54E+08	6.60E-06
off/on	-0.01	0.22
		-11.155581

S 2. Calculated drop of thermal diffusivities from heavy ion irradiation (top values) and proton irradiation (bottom values). Thermal diffusivities drop ratio between heavy ion and protons is shown in the lower right corner. Excel file available in the repository of this paper [3].

Both tables are available in an Excel file on the repository for this paper [23].